

AI Network Failure Prediction System

DAILY REPORT

Generated: 13 September 2026 22:05:36

1. Executive Summary

Metric	Value
Overall Network Health	WARNING
SLA Percentage	100.0%
Uptime Records	89
Downtime Records	0
Report Type	Daily
Algorithm Used	XGBoost + SHAP
Devices Monitored	Router, Switch, Firewall

2. Device Status

Device	Status	Health
Router	■ Healthy	Good
Switch	■■ Warning	Check Required
Firewall	■■ Warning	Check Required

3. Failure Breakdown by Device

Device	Failure Alerts
Firewall	19
Router	2
Switch	5

4. Recent Alerts (Last 10)

Timestamp	Device	Severity	Lead Time
2026-09-13 22:04:42	Switch	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:42	Firewall	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:40	Firewall	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:38	Firewall	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:37	Switch	MEDIUM	Near-term (10–30 seconds)

2026-09-13 22:04:36	Firewall	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:35	Switch	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:34	Firewall	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:33	Switch	MEDIUM	Near-term (10–30 seconds)
2026-09-13 22:04:32	Switch	MEDIUM	Near-term (10–30 seconds)

5. AI Model Performance

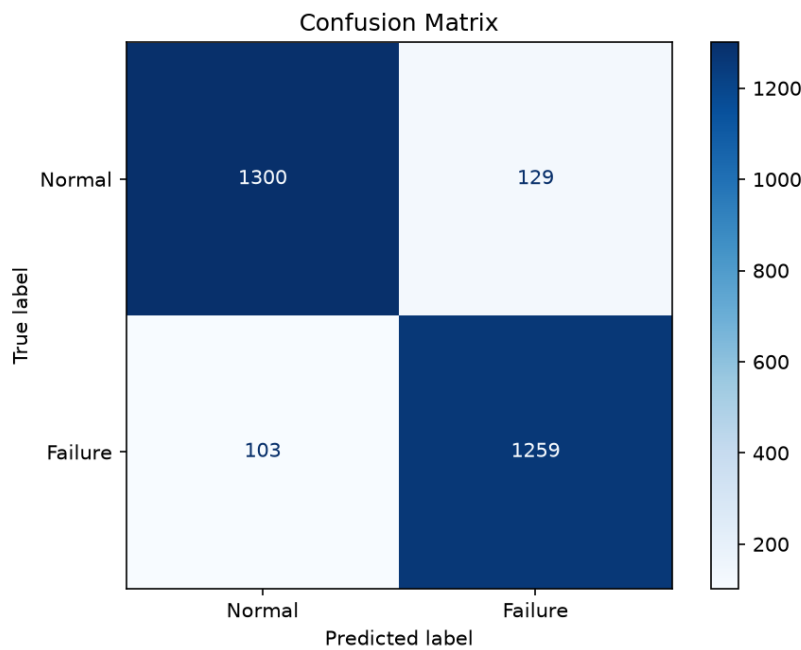
The XGBoost model was trained with SMOTE balancing on 9,000 live-generated readings across 3 devices. Evaluation charts (confusion_matrix.png, roc_curve.png, shap_summary_bar.png, shap_beeswarm.png) are available in the project folder.

Metric	Description
Accuracy	Overall correct predictions (typical: 93-96%)
Precision	When FAILURE predicted, how often it is correct
Recall	Of all real failures, how many were caught
F1-Score	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve (typical: 0.97-0.99)

6. SHAP Explainability

SHAP (SHapley Additive exPlanations) is integrated to explain every individual failure prediction. For each alert, the system shows how much each metric (CPU%, Memory%, Latency, Errors, Traffic) contributed to the AI's decision. This makes the system fully transparent and actionable for network administrators.

Confusion Matrix:



7. SLA & Uptime Statistics

Metric	Value
SLA Percentage	100.0%
Total Records	89
Uptime Records	89
Downtime Records	0
Grade	Excellent

Generated by: Explainable AI-Based Predictive Failure Detection for Network Devices Using XGBoost and SHAP